LaTeX-Workshop

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- What is LaTeX?
- What is it used for?
- What is it NOT?
- Why do we need to use it?

Syntax

- Designed to focus on writing
- Text will be output as text
  - White space is ignored (including newlines)
  - Empty line signifies end of paragraph
  - LaTeX automagically does hyphenation and outlining
  - "Magical" characters:
    # $ % ^ & _ { } ~ \
  - Cannot use these in running text
- Comments start with a %

Syntax

- Commands start with a \:
  `\somecommand[optional]{required}`
- Groups start with { and end with }
  - Useful if command influences all following text
    `This {\bf text is bold} but this it not`
    This **text is bold** but this it not
- Environments start with `\begin{envname}` and end with `\end{envname}`
  - Define new commands that only work in environment
  - Change flow of text
  - Introduce special elements (i.e. non-text) in document

Editors

- There exist several LaTeX editors
- We generally recommend using one of the following editors below
- Online editor:
  - Overleaf (`www.overleaf.com`) **recommended**
- Offline editors (all cross platform):
  - Texmaker (`www.xm1math.net/texmaker`)
  - TeXstudio (`www.texstudio.org`)
  - TeXworks (`www.tug.org/texworks`)

Overleaf

- If you're not familiar with LaTeXyet, we recommend you to use Overleaf during this tutorial
- You can register an account or login with your Google account at `www.overleaf.com`

Document Structure

- Document consists of two parts:
  - Preamble
  - Content
- Further forces LaTeX's focus on content

The preamble

- Set up for the document
  - Documentclass
  - Loading packages
  - Document wide settings
  - New commands/environments
  - Meta information (title, author, etc. . .)
- Allows us to change anything about the document
  - Also paper size, margins, etc
  - After preamble no longer possible

The document class

- Every document has document class
- Specifies general settings like paper size, margins, headers, footers, etc. . .
- Is set by `\documentclass` command:
  `\documentclass[a4paper, 10pt]{article}`
- Different types of document classes:
  - article
  - beamer
  - report
  - book
  - letter
  - cover-article
  - . . .

Importing packages

- LaTeX files with commands/environments
- Can also be used to set setting (i.e. geometry)
- Extends LaTeX's usual capabilities.
- Imported with \usepackage
  `\usepackage[hidelinks]{hyperref}`
- Contains all kinds of stuff:
  - `hyperref` for making hyperlinks
  - `graphicx` for importing images
  - `amsmath` for even better math support
  - `colorx` for changing colors
  - `tikz` for drawing figures in LaTeX
  - `coffee` for adding coffee stains
  - `enigma` for confusing the allied forces

Setting settings

- Settings depend on packages used and needs of author
- `\title{}`, `\author{}` and `\date{}` are quite common

The content

- Content goes into document environment:

```
\begin{document}
    ...
\end{document}
```

- Normally starts with a title: `\maketitle`
- For longer documents a table of contents: `\tableofcontents`

Document structure

- Layout is done by LaTeX, structure is important
- Different kinds of headings:
  - `\chapter` only in book and report
  - `\section`
  - `\subsection`
  - `\subsubsection`
  - `\paragraph` typeset in running text
  - `\subparagraph`
  - Also a starred version to disable numbering: `\section*{}`
- `\appendix` changes number to appendix numbering

Text flow

- LaTeX ignores white space
- Works by splitting text up into tokens and paragraphs
- Tokens are separated by whitespace or punctuation
- Empty line ends paragraph
- Sometimes you want to force a space or newline
- For space, the non-breaking space: ~
- For newline: \\
  - Warning! This does *not* start a new paragraph

Simple formatting

- Changing text size:
  - `\tiny`
  - `\scriptsize`
  - `\footnotesize`
  - `\small`
  - `\normalsize`
  - `\large`
  - `\Large`
  - `\LARGE`
  - `\huge`
  - `\Huge`

- **Bold text:**
  `\textbf{text} {\bf text}`
- *Italic text:*
  `\textit{text} {\it text}`
- *Emphasised text:*
  `\emph{text}`
- <u>Underlined text:</u>
  `\underline{text}`
- `Monotype text:`
  `\texttt{text} {\tt text}`

Lists

- Three kinds of list:
  1. `itemize` bulleted list like the outer list
  2. `enumerate` numbered list like this one
  3. `description` list of definitions
- All lists are environments
- New items are added with `\item`
- Lists can be nested up to five levels deep
- Numbering happens automatically

Referencing

- LaTeX numbers a lot of stuff
- Referring back to parts of document is easy
- Name object with `\label{name}`
- Get number with `\ref{name}`
- Get the page number of object with `\pageref{name}`
- (With `hyperref`) Get name, number, and link `\autoref{name}`
- Can be used to refer to:
  - Sections, subsections, etc...
  - Items in an enumeration
  - Equations
  - Figures
  - Tables

What are floats?

- LaTeX tries to structure text for optimal flow
- Does not work if we include figures and tables
- Solution: Floats
- Float is a container that cannot be broken over pages
- LaTeX moves floats through document for optimal placement
- Works good for images, tables, pseudocode, and other blocks
- Allows us to enrich content
    - Captions
    - Subfigures
- Can also define our own floats

Controlling float placement

- LaTeX tends to move floats all over the place
- Two ways to control floats:
  - Placement specifiers
  - Float barriers
- For reports: pictures at end is fine, lots of empty space is not

Controlling float placement

- LaTeX tends to move floats all over the place
- Two ways to control floats:
  - Placement specifiers
    - Optional argument to the float
    - Does not always work "correctly"
    - Pictures still float
    - `h` I want my float here
    - `H` Force float here (requires `float` package)
    - `t` I want my float at the top of the page
    - `b` I want my float at the bottom of the page
    - `!` I really want this
    - `h!` tells LaTeX you really want the float here
  - Float barriers
- For reports: pictures at end is fine, lots of empty space is not

Controlling float placement

- LaTeX tends to move floats all over the place
- Two ways to control floats:
  - Placement specifiers
  - Float barriers
    - Stops floats from moving past a point
    - Useful to keep floats within section
    - Require the `placeins` package
    - Created with command `\FloatBarrier`
    - Automatic float barriers at sections: `\usepackage[section]{placeins}`
- For reports: pictures at end is fine, lots of empty space is not

Working with images

```
\begin{figure}[htb]
    \includegraphics[width=\textwidth]{path/to/file}
    \caption{Something sensible about this image}
    \label{fig:my_image}
\end{figure}
```

- [htb] is placement specifier
- `\includegraphics` for importing of the image
- `\caption` for text below image and numbering
- `\label` so we can refer later

Working with tables

```latex
\begin{table}
    \caption{Caption above table}
    \label{tbl:square}
    \begin{tabular}{|c|c|c|}
        \hline
        2 & 7 & 6 \\
        \hline
        9 & 5 & 1 \\
        \hline
        4 & 3 & 8 \\
        \hline
    \end{tabular}
\end{table}
```

Table: Caption above table

2	7	6
9	5	1
4	3	8

Three useful sites:

- `en.wikibooks.org/wiki/LaTeX/Tables`
- `www.latex-tables.com`
- `www.tablesgenerator.com`

Working with floats

- LaTeX knows the best way to position floats
- Bunch of pictures at the end of report not bad
  - Graders know why
  - Large sections of white space are more annoying
- For figures: changing the width so two pictures fit on page
- Text on both sides of float help with positioning
  - More text $\Rightarrow$ more things for LaTeX to play with
- Float barriers can help if document is broken up in logical sections
  - E.G. all the figures for exercise 1 stay with exercise 1

Mathematics

- Most of the following commands need the package amsmath
- `$ ... $` or `\( ... \)` short hand for inline math
- `$$ ... $$` or `\[ ... \]` short hand for display math mode
- The equation environment displays numbered equations
- The align environment is used to arrange equations over multiple lines. `\\` denotes a new line and & indicates a point where the equations should be aligned
- Spaces are derived from the expressions, `$2-3$` and `$2 - 3$` both print to $2 - 3$

Common commands

- `\times` is $\times$, `\cdot` is $\cdot$
- `\sqrt{a}` is $\sqrt{a}$
- `\frac{a}{b}` is $\frac{a}{b}$
- `a^{b}` is a^{b}
- `a_{b}` is a_{b}
- All greek letters are available. `\Delta` is Δ and `\delta` is δ
- Text can be added to the equation with `\text{}`.
  `2 - 3 \text{ hallo } 3 \times 4` is $2 - 3 \text{ hallo } 3 \times 4$

Operators

- `\cos (2\pi)` is $\cos(2\pi)$
- `\sin^2 (2\theta)` is $\sin^2(2\theta)$
- `\tan (2)` is $\tan(2)$
- `\lim_{x \to \infty} \exp(-x) = 0` is $\lim_{x \to \infty} \exp(-x) = 0$
- `x \equiv a \pmod{b}` is $x \equiv a \pmod{b}$
- https://en.wikibooks.org/wiki/LaTeX/Mathematics lists more commands
- detexify.kirelabs.org can identify a latex symbol from a drawing

Why chemists use TeX

- Chemistry writing needs consistent notation for formulas, reactions, units, and references
- LaTeX is strong at keeping that notation uniform across long reports and theses
- Two packages show up all the time in chemistry documents:
  - `mhchem` for chemical formulas and reactions
  - `siunitx` for numbers, units, and uncertainties
- The goal is readable source and publication-quality output

Chemical formulas and reactions

- `\ce` formats formulas, ions, and reactions
- Examples:
  - Water: `H2O`: H_2O
  - Oxidation state: `Fe\^{}\{2+\}`: Fe^{2+}
  - Reaction: `2 H2 + O2 -> 2 H2O`: $2\,H_2 + O_2 \longrightarrow 2\,H_2O$
  - Equilibrium: `CH3COOH + H2O <=> CH3COO\^{}- + H3O+`: $CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$
- This keeps subscripts, charges, arrows, and stoichiometry consistent

Units, quantities, and data

- `siunitx` keeps units separated from values
- Examples:
  - Volume: `\SI{25}{\milli\liter}`: 25 mL
  - Temperature: `\SI{298.15}{\kelvin}`: 298.15 K
  - Concentration: `\SI{1.20e-3}{\mole\per\liter}`: $1.20 \times 10^{-3}\,\mathrm{mol\,L^{-1}}$
  - Count with formatting: `\num{6.022e23}`: 6.022×10^{23}
- Useful for lab reports, tables, and plots where significant figures matter

Common chemistry document tips

- Put reaction schemes, spectra, and tables in floats so they can be captioned and referenced
- Use `\label` and `\ref` for compounds, schemes, and figures
- Keep citations in a `.bib` file so experimental sources and papers stay manageable
- If your report has a lot of notation, define a few small commands in the preamble to reduce repetition

Source Code

- LaTeX can be used to format source code
  - Requires `listings` package
- Can include (parts of) document, snippets, and lines
- Automatic syntax recognition
- Allows different styles
  - Most languages have settings online
- `\lstinputlisting{file}` to include file
- `\begin{lstlisting}` and `\end{lstlisting}` for snippets
- `\lstinline{code}` for inline code

Settings

`listings` offers several settings for each command, can be set globally using `\lstset{}`

Setting	Values	Explanation
breaklines	true/false	If code should be broken over lines
breakatwhitespace	true/false	If code can only break at whitespace
language	string	The language the code is written in
numbers	none/left/right	Where to position the line numbers
stepnumber	number	The numbers in between two steps

Bibtex

Useful package for referencing sources.

- Stores references in seperate ".bib" file.
  - Set style with `\bibliographystyle{plain}`
  - Include bib file with `\bibliography{filename}`
  - These should be placed in this order at the location you want the reference list to be printed
- `\cite{}` to cite the refrerences in the bib file
- More information at https://en.wikibooks.org/wiki/LaTeX/Bibliography_Management.

General

Package	Description
`enumitem`	Changes the symbols used in enumerations
`babel`	For writing text in languages that are not English
`fancyhdr`	For making headers and footers
`tikz`	For the creation of drawings in LaTeX
`moderncv`	For making a résumé

Linguistics

Package	Description
`tipa`	For symbols from the IPA
`qtree`	For drawing (syntactic) trees
`tikz-qtree`	For drawing (syntactic) trees with `tikz`
`gb4e`	For glosses and the likes
`phonrule`	For phonological rules
`lplfitch`	For typesetting natural deduction proofs

Let's Pracice!

- We have prepared a template for you to practice with
- You can find it on Overleaf at `t.ly/iyYsG`
- We will go through the exercises together, but feel free to ask questions and experiment with the code!